* 坐标表象本征函数的正交性和完备性

$$\hat{A}|a_i\rangle = a_i|a_i\rangle ,$$

令 $\langle x|a_i\rangle \equiv u_i(x)$,

(1) 正交性： 由 $\langle a_j|a_i\rangle = \delta_{ij}$

$$\Rightarrow \int dx \langle a_j|x\rangle\langle x|a_i\rangle = \delta_{ij}$$

$$\Rightarrow \boxed{\int dx\, u_j^*(x) u_i(x) = \delta_{ij}}$$

(2) 完备性： 由 $\sum_i |a_i\rangle\langle a_i| = \mathbb{1}$ ，左乘 $\langle x'|$，右乘 $|x\rangle$

$$\Rightarrow \sum_i \langle x'|a_i\rangle\langle a_i|x\rangle = \langle x'|x\rangle = \delta(x-x')$$

$$\Rightarrow \boxed{\sum_i u_i^*(x') u_i(x) = \delta(x-x')}$$

8 坐标表象下，一维线性谐振子系统的薛定谔方程.

· $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$,

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2$, 左乘 $\langle x|$

$\Rightarrow$ $i\hbar\frac{\partial}{\partial t}\langle x|\psi(t)\rangle = \langle x|\hat{H}|\psi(t)\rangle$

$\langle x|\psi(t)\rangle \equiv \psi(x,t)$ (波函数) .

左边: $i\hbar\frac{\partial}{\partial t}\psi(x,t)$.

右边: $\langle x|\left[\frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2\right]|\psi(t)\rangle$,

· 第一项: $\langle x|\frac{\hat{p}^2}{2m}|\psi(t)\rangle = \frac{1}{2m}\int dx'\langle x|\hat{p}^2|x'\rangle\langle x'|\psi(t)\rangle$

$= \frac{1}{2m}\int dx'\left[(-i\hbar\frac{\partial}{\partial x})^2\delta(x-x')\right]\psi(x't)$

$$= \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)^2 \int dx'\, \delta(x-x')\,\psi(x't)$$

$$= \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)^2 \psi(x,t) = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(xt) .$$

· 对 $\frac{1}{2}kx^2$: $\frac{m\omega^2}{2}\langle x|\hat{x}^2|\psi(t)\rangle$

$$= \frac{m\omega^2}{2}x^2\langle x|\psi(t)\rangle = \frac{m\omega^2}{2}x^2\psi(xt) .$$

$$\Rightarrow \quad \underline{i\hbar\frac{\partial}{\partial t}\psi(x,t) = \left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}x^2\right]\psi(xt)} .$$

$\hat{H}$ 的本征值方程的坐标表象.

$$\underline{\hat{H}|E\rangle = E|E\rangle} ,$$

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2$. 左乘 $\langle x|$

$$\Rightarrow \quad \langle x|\hat{H}|E\rangle = E\langle x|E\rangle$$

· $\langle x|E\rangle \equiv \psi_E(x)$

· 左方：$\langle x|\hat{H}|E\rangle = \int dx' \langle x|\hat{H}|x'\rangle\langle x'|E\rangle$

$= \int dx' \langle x| \left[\frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2\right]|x'\rangle \psi_E(x')$

$= \int dx' \left[\frac{1}{2m}(-i\hbar\frac{\partial}{\partial x})^2 + \frac{m\omega^2}{2}x^2\right]\delta(x-x')\psi_E(x')$

$= \left[\frac{1}{2m}(-i\hbar\frac{\partial}{\partial x})^2 + \frac{m\omega^2}{2}x^2\right]\int dx' \delta(x-x')\psi_E(x')$

$= \left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{m}{2}\omega^2 x^2\right]\psi_E(x)$

· 右方：$E\,\psi_E(x)$.

$\Rightarrow$ $\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2}x^2\right]\psi_E(x) = E\,\psi_E(x)$

解：

$$\begin{cases} E = E_n = (n+\frac{1}{2})\hbar\omega, & n = 0, 1, 2, \cdots \\ \psi_E(x) = \psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n(\xi)\, e^{-\xi^2/2} \\ H_n(\xi) \equiv e^{\xi^2/2}\left(\xi - \frac{d}{d\xi}\right)^n e^{-\xi^2/2} \quad (\text{厄米多项式}) \\ \xi \equiv x/x_0, \quad x_0 \equiv \sqrt{\frac{\hbar}{m\omega}} \end{cases}$$

(B) 动量表象

- 动量算符 $\hat{p}$ 的本征值方程：

(一维) $\hat{p}|p\rangle = p|p\rangle$. $(-\infty < p < \infty)$

正交性： $\langle p'|p\rangle = \delta(p-p')$

完备性： $\int_{-\infty}^{\infty} dp\ |p\rangle\langle p| = \mathbb{1}$

- 动量表象中态矢量 $|\psi(t)\rangle$ 的表示

$$|\psi(t)\rangle = \mathbb{1}\,|\psi(t)\rangle = \int_{-\infty}^{\infty} dp\ |p\rangle\langle p|\psi(t)\rangle$$

$$= \int_{-\infty}^{\infty} dp\ |p\rangle\ \underbrace{\psi(p,t)}_{\text{展开系数}}$$

其中，$\psi(p,t) \equiv \langle p|\psi(t)\rangle$ 为态矢量 $|\psi(t)\rangle$ 的动量表象，称为动量空间波函数，$|\psi(p,t)|^2$ 给出 t 时刻粒子动量为 p 的概率。

↓

· 动量表象下算符的表示

$$\hat{F} = \mathbb{1}\hat{F}\mathbb{1} = \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' |p\rangle\langle p|\hat{F}|p'\rangle\langle p'|$$

$$= \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' F_{pp'} |p\rangle\langle p'|$$

其中 $F_{pp'} \equiv \langle p|\hat{F}|p'\rangle$ （矩阵元）

· 动量表象下坐标算符的矩阵元

$$x_{pp'} \equiv \langle p|\hat{x}|p'\rangle = i\hbar\frac{\partial}{\partial p}\delta(p-p')$$

（证明略）

① 算符 $\hat{x}^n$ 的矩阵元

$$x^n_{pp'} = \langle p|(\hat{x})^n|p'\rangle = (i\hbar\frac{\partial}{\partial p})^n \delta(p-p')$$

② 动量表象下，算符 $\hat{p}^n$ 的矩阵元

$$P^n_{pp'} \equiv \langle p|\hat{p}^n|p'\rangle = p^n\delta(p-p')$$

* 坐标表象和动量表象间波函数与矩阵元的相互关系

- $\psi(x) \equiv \langle x|\psi\rangle = \int_{-\infty}^{\infty} dp\ \langle x|p\rangle\langle p|\psi\rangle$

$$= \int_{-\infty}^{\infty} dp\ \langle x|p\rangle\ \psi(p)$$

$\Rightarrow$ $\hat{p}|p\rangle = p|p\rangle$ ，左乘 $\langle x|$

$\Rightarrow$ $\langle x|\hat{p}|p\rangle = p\langle x|p\rangle$ 。

$\Rightarrow$ $\int dx'\langle x|\hat{p}|x'\rangle\langle x'|p\rangle = p\langle x|p\rangle$

$$\Rightarrow \quad \int dx' \left(-i\hbar\frac{\partial}{\partial x}\delta(x-x')\right)\langle x'|p\rangle = p\langle x|p\rangle$$

$$\Rightarrow \quad -i\hbar\frac{\partial}{\partial x}\int dx'\,\delta(x-x')\langle x'|p\rangle = p\langle x|p\rangle$$

$$\Rightarrow \quad -i\hbar\frac{\partial}{\partial x}\langle x|p\rangle = p\langle x|p\rangle$$

解：

$$\Rightarrow \quad \langle x|p\rangle = A\exp\left[\frac{ipx}{\hbar}\right]$$

$$\left(\quad A=\frac{1}{\sqrt{2\pi\hbar}}\quad\right) \quad ? \quad \begin{cases}\psi(x)\to\psi(x,t)\\ \psi(p)\to\psi(p,t)\end{cases}$$

$$\Rightarrow \quad \boxed{\psi(x,t)=\int_{\omega}^{\omega} dp\,\frac{1}{\sqrt{2\pi\hbar}}\exp\left(\frac{ipx}{\hbar}\right)\psi(p,t)}$$

F.T.

· 确定 A 值.

$$\langle p'|p\rangle = \delta(p'-p)\ .$$

左:
$$\langle p'|p\rangle = \langle p'|\mathbb{1}|p\rangle$$

$$= \int_{-\infty}^{\infty} dx\ \langle p'|x\rangle\langle x|p\rangle$$

$$= |A|^2 \int_{-\infty}^{\infty} dx\ e^{-\frac{ip'x}{\hbar}}\ e^{\frac{ipx}{\hbar}}$$

$$= |A|^2 \int_{-\infty}^{\infty} dx\ e^{i(p-p')x/\hbar}$$

$$= |A|^2 \hbar \int_{-\infty}^{\infty} dx\ e^{i(p-p')x}$$

$$= |A|^2 2\pi\hbar\ \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\ e^{i(p-p')x}$$

$$= |A|^2 2\pi\hbar\ \delta(p'-p) \quad \Rightarrow |A|^2 2\pi\hbar = 1$$

$$\Rightarrow A = \frac{1}{\sqrt{2\pi\hbar}} \Rightarrow \text{(平面波)}^{\text{一维}}$$

* 对易关系

在坐标表象中，

$[\hat{x}, \hat{p}] = i\hbar \Rightarrow [x, -i\hbar\frac{\partial}{\partial x}] = i\hbar$。

可证明，在动量表象中，

$[\hat{x}, \hat{p}] = i\hbar \Rightarrow [i\hbar\frac{\partial}{\partial p}, p] = i\hbar$。

3、表象变换 （分立谱）

A-表象 (n维)	B-表象 (n维)
• $\hat{A}\|a_i\rangle = a_i\|a_i\rangle$	• $\hat{B}\|b_i\rangle = b_i\|b_i\rangle$
• $\|\psi(t)\rangle \Rightarrow \psi_A = \begin{pmatrix} \langle a_1\|\psi\rangle \\ \langle a_2\|\psi\rangle \\ \vdots \\ \langle a_n\|\psi\rangle \end{pmatrix}$	• $\|\psi(t)\rangle \Rightarrow \psi_B = \begin{pmatrix} \langle b_1\|\psi\rangle \\ \langle b_2\|\psi\rangle \\ \vdots \\ \langle b_n\|\psi\rangle \end{pmatrix}$
• $\hat{F} \Rightarrow F_A = \begin{pmatrix} \vdots & & \\ \cdots & F^A_{ij} & \cdots \\ \vdots & & \end{pmatrix}$	• $\hat{F} \Rightarrow F_B = \begin{pmatrix} \vdots & & \\ \cdots & F^B_{ij} & \cdots \\ \vdots & & \end{pmatrix}$
（基矢 $\{\|a_i\rangle\}$ 正交、完备）	（基矢 $\{\|b_j\rangle\}$ 正交、完备）

表象变换：$\begin{cases} \psi_A \longleftrightarrow \psi_B \\ F_A \longleftrightarrow F_B \end{cases}$

(a) 基矢间变换

引入算符：

$$\begin{cases} \hat{S} \equiv \sum_k |a_k\rangle\langle b_k| \\ \hat{S}^+ \equiv \sum_k |b_k\rangle\langle a_k| \end{cases}$$

基矢间变换：

- $\hat{S}|b_j\rangle = \sum_k |a_k\rangle\langle b_k|b_j\rangle$

$$= \sum_k |a_k\rangle \delta_{kj} = |a_j\rangle$$

- $\hat{S}^+|a_i\rangle = \sum_k |b_k\rangle\langle a_k|a_i\rangle$

$$= \sum_k |b_k\rangle \delta_{ki} = |b_i\rangle$$

- $\hat{S}$ 是幺正算符 $(\hat{S}^+ = \hat{S}^{-1})$

证明：

$$\hat{S}\hat{S}^{\dagger} = \left(\sum_k |a_k\rangle\langle b_k|\right)\left(\sum_{k'} |b_{k'}\rangle\langle a_{k'}|\right)$$

$$= \sum_k \sum_{k'} |a_k\rangle\langle b_k|b_{k'}\rangle\langle a_{k'}|$$

$$= \sum_k \sum_{k'} |a_k\rangle \delta_{kk'} \langle a_{k'}|$$

$$= \sum_k |a_k\rangle\langle a_k| = \mathbb{1}$$

类似，可得：$\hat{S}^{\dagger}\hat{S} = \mathbb{1}$

$$\hat{S}\hat{S}^{\dagger} = \hat{S}^{\dagger}\hat{S} = \mathbb{1} \Rightarrow \hat{S}^{\dagger} = \hat{S}^{-1}$$ 证毕.

- $\hat{S}$ 的 A-表象与 B-表象.

A-表象：$S_{ij}^{A} = \langle a_i|\hat{S}|a_j\rangle$

$$= \sum_k \langle a_i|a_k\rangle\langle b_k|a_j\rangle = \sum_k \delta_{ik}\langle b_k|a_j\rangle$$

$$= \langle b_i|a_j\rangle .$$

B-表象：$S_{ij}^{B} = \langle b_i | \hat{S} | b_j \rangle$

$$= \sum_k \langle b_i | a_k \rangle \langle b_k | b_j \rangle$$

$$= \sum_k \langle b_i | a_k \rangle \delta_{kj}$$

$$= \langle b_i | a_j \rangle$$

即 $S_{ij}^{A} = S_{ij}^{B} = \langle b_i | a_j \rangle \equiv S_{ij}$

$$\begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & & \\ S_{n1} & \cdots & S_{nn} \end{pmatrix} \equiv S \quad (\text{变换矩阵})$$

(b) 态矢量的表象变换

$$\psi_A = \begin{pmatrix} \langle a_1|\psi\rangle \\ \vdots \\ \langle a_n|\psi\rangle \end{pmatrix}, \quad \psi_B = \begin{pmatrix} \langle b_1|\psi\rangle \\ \vdots \\ \langle b_n|\psi\rangle \end{pmatrix}$$

$\psi_A \to \psi_B$

$$\langle b_i|\psi\rangle = \sum_j \langle b_i|a_j\rangle\langle a_j|\psi\rangle$$

$$= \sum_j S_{ij} \langle a_j|\psi\rangle \quad (i=1,2,\cdots n)$$

$$\Rightarrow \begin{pmatrix} \langle b_1|\psi\rangle \\ \vdots \\ \langle b_n|\psi\rangle \end{pmatrix} = \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & & \\ S_{n1} & \cdots & S_{nn} \end{pmatrix} \begin{pmatrix} \langle a_1|\psi\rangle \\ \vdots \\ \langle a_n|\psi\rangle \end{pmatrix}.$$

即 $\boxed{\psi_B = S\,\psi_A}$

$$S^+\psi_B = S^+S\psi_A = \psi_A$$

$$\Rightarrow \qquad \psi_B \to \psi_A$$

$$\boxed{\psi_A = S^+\psi_B}$$

(C) 算符矩阵的表象变换

$$F^A = \begin{bmatrix} F^A_{11} & \cdots & F^A_{1n} \\ \vdots & & \\ F^A_{1n} & \cdots & F^A_{nn} \end{bmatrix}$$

$$(F^A_{ij} = \langle a_i|\hat{F}|a_j\rangle)$$

$$F^B = \begin{bmatrix} F^B_{11} & \cdots & F^B_{1n} \\ \vdots & & \\ F^B_{1n} & \cdots & F^B_{nn} \end{bmatrix} \qquad (F^B_{ij} = \langle b_i|\hat{F}|b_j\rangle)$$

- $F^A \to F^B$

$$F^B_{ij} = \langle b_i | \hat{F} | b_j \rangle$$

$$= \sum_{kl} \langle b_i | a_k \rangle \langle a_k | \hat{F} | a_l \rangle \langle a_l | b_j \rangle$$

$$= \sum_{kl} S_{ik} F^A_{kl} \langle b_j | a_l \rangle^*$$

$$= \sum_{kl} S_{ik} F^A_{kl} S^*_{jl}$$

$$= \sum_{kl} S_{ik} F^A_{kl} S^+_{lj} \qquad (i, j = 1, 2, \ldots n)$$

$$\Rightarrow \boxed{F^B = S F^A S^+}$$

- $F^B \to F^A$

$$\boxed{F^A = S^+ F^B S}$$

(d) 幺正变换的若干性质

(I) 保持本征谱不变

$$F_A \phi_A^{(i)} = \lambda_A^{(i)} \phi_A^{(i)}$$ ，两边左乘 S，

$$\Rightarrow \quad S F_A S^{+} S \phi_A^{(i)} = \lambda_A^{(i)} S \phi_A^{(i)}$$

$$\Rightarrow \quad F_B \phi_B^{(i)} = \lambda_A^{(i)} \phi_B^{(i)}$$

(II) 体持算符的厄米性不变 $\left(SFS^{+}\right)^{\dagger}=SFS^{+}$

(III) 体持算符的对易或为对易关系不变

设 $[F,G]_{\pm}=C$, (C为常数)

(IV) 体持量值 $\langle\varphi|\hat{F}|\psi\rangle$ $(\varphi^{+}F\psi)$

不变

· 令 $\hat{F}=\mathbb{1}$ $\Rightarrow$ $\langle\varphi|\psi\rangle$ 不变！

内积

· 令 $|\varphi\rangle=|\psi\rangle$ $\Rightarrow$ $\langle\psi|\hat{F}|\psi\rangle$ 不变！

平均值

· 令 $\hat{F}=\mathbb{1}$, $|\varphi\rangle=|\psi\rangle$ $\Rightarrow$ $\langle\psi|\psi\rangle$

归一化不变！

(V) 保持算符矩阵的迹不变

$$tr[\hat{F}^A] = tr[S^\dagger S \hat{F}^A] = tr[S \hat{F}^A S^\dagger] = tr[\hat{F}^B]$$

$$\Rightarrow \quad \sum_n \langle a_n | \hat{F} | a_n \rangle = \sum_m \langle b_m | \hat{F} | b_m \rangle \equiv tr[\hat{F}]$$

- $tr[|a_n\rangle\langle a_m|] = \delta_{n,m}$,
- $tr[|b_m\rangle\langle a_m|] = \langle a_m | b_m \rangle$

(VI) 保持算符矩阵的行列式不变

$$\begin{aligned} \det[\hat{F}^A] &= \det[S^\dagger \hat{F}^B S] \\ &= \det[S^\dagger] \det[\hat{F}^B] \det[S] \\ &= \det[S^\dagger S] \det[\hat{F}^B] \\ &= \det[\mathbb{1}] \det[\hat{F}^B] \\ &= \det[\hat{F}^B] \end{aligned}$$

坐标表象与动量表象间的变换

$$\psi(x,t) = \int_{-\infty}^{\infty} dp \frac{1}{\sqrt{2\pi\hbar}} \exp\left[\frac{i}{\hbar} px\right] \psi(p,t) ,$$

$$\psi(p,t) = \int_{-\infty}^{\infty} dx \frac{1}{\sqrt{2\pi\hbar}} \exp\left[-\frac{i}{\hbar} px\right] \psi(x,t) ,$$

$$F_{x'x''} = \int_{-\infty}^{\infty} dp' dp'' \frac{1}{2\pi\hbar} \exp\left[\frac{i}{\hbar}(p'x' - p''x'')\right] F_{p'p''} ,$$

$$F_{p'p''} = \int_{-\infty}^{\infty} dx' dx'' \frac{1}{2\pi\hbar} \exp\left[-\frac{i}{\hbar}(p'x' - p''x'')\right] F_{x'x''} .$$

其中. $F_{x'x''} \equiv \langle x'|\hat{F}|x''\rangle$.

$F_{p'p''} \equiv \langle p'|\hat{F}|p''\rangle$.

例：已知 $\chi_{x'x''} = \langle x'|\hat{\chi}|x''\rangle$

$= x'\delta(x'-x'')$

求 $\chi_{p'p''} = \langle p'|\hat{\chi}|p''\rangle = ?$

$\hat{F} \to \hat{\chi}$,

$$\chi_{p'p''} = \int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dx'' \frac{1}{2\pi\hbar} \exp\left[-\frac{i}{\hbar}(p'x'-p''x'')\right] x'\delta(x'-x'')$$

$$= \int_{-\infty}^{\infty} dx' \frac{1}{2\pi\hbar} \exp\left[-\frac{i}{\hbar}(p'x'-p''x')\right] x'$$

$$= \int_{-\infty}^{\infty} dx' \frac{1}{2\pi\hbar} (i\hbar\frac{\partial}{\partial p'}) \exp\left[-\frac{i}{\hbar}(p'-p'')x'\right]$$

$$= (i\hbar\frac{\partial}{\partial p'}) \int_{-\infty}^{\infty} dx' \frac{1}{2\pi\hbar} \exp\left[-\frac{i}{\hbar}(p'-p'')x'\right]$$

$$= i\hbar\frac{\partial}{\partial p'}\delta(p'-p'') \quad . \qquad \text{ok!}$$

4 能量表象与动量表象间的变换.

例：一维线性谐振子系统

能量表象：$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$.

$$\hat{H}|E_n\rangle = E_n|E_n\rangle \quad , \quad \begin{cases} \langle E_{n'}|E_n\rangle = \delta_{n'n} \\ \sum_{n=0}^{\infty} |E_n\rangle\langle E_n| = \mathbb{1} \end{cases}$$

$$|\psi(t)\rangle = \mathbb{1}|\psi(t)\rangle = \sum_{n=0}^{\infty} |E_n\rangle\langle E_n|\psi(t)\rangle$$

$$= \sum_{n=0}^{\infty} |E_n\rangle \psi_{E_n}(t)$$

能量表象中的波函数.

$$\cdot \ \psi(t) = \begin{pmatrix} \langle E_1|\psi(t)\rangle \\ \langle E_2|\psi(t)\rangle \\ \vdots \end{pmatrix} = \begin{pmatrix} \psi_{E_1}(t) \\ \psi_{E_2}(t) \\ \vdots \end{pmatrix}$$ 态的矩阵

动量表象：$\hat{p}|p\rangle = p|p\rangle \quad . \ -\infty < p < \infty$

$$\begin{cases} \langle p'|p\rangle = \delta(p'-p) \\ \int dp \, |p\rangle\langle p| = \mathbb{1} \end{cases}$$

$$|\psi(t)\rangle = \mathbb{1}|\psi(t)\rangle = \int dp\, |p\rangle\langle p|\psi(t)\rangle$$

$$= \int dp\, |p\rangle \underline{\psi(p,t)}$$

动量表象中的波函数

$$\psi(t) = \begin{pmatrix} \vdots \\ \psi(pt) \\ \vdots \end{pmatrix}$$

态的矢量.

- 态间变换

(a) $\psi_{E_n}(t) \rightarrow \psi(p,t)$

$$\underline{\psi_{E_n}(t)} = \langle E_n|\psi(t)\rangle = \int dp\, \langle E_n|p\rangle\langle p|\psi(t)\rangle$$

$$= \int dp\, \langle E_n|p\rangle \underline{\psi(pt)}$$

(b) $\psi(pt) \rightarrow \psi_{E_n}(t)$

$$\underline{\psi(pt)} = \langle p|\psi(t)\rangle = \sum_n \langle p|E_n\rangle\langle E_n|\psi(t)\rangle$$

$$= \sum_n \langle p|E_n\rangle \psi_{E_n}(t)$$

$$= \sum_n \langle E_n|p\rangle^* \underline{\psi_{E_n}(t)}.$$

- 矩阵间变换

$$F_{nn'} = \langle E_n | \hat{F} | E_{n'} \rangle \qquad (\text{能量表象})$$

$$F_{pp'} = \langle p | \hat{F} | p' \rangle \qquad (\text{动量表象})$$

(a) $F_{nn'} \to F_{pp'}$

$$\underline{F_{nn'}} = \langle E_n | \hat{F} | E_{n'} \rangle$$

$$= \int dp\, dp' \langle E_n | p \rangle \langle p | \hat{F} | p' \rangle \langle p' | E_{n'} \rangle$$

$$= \int dp\, dp' \langle E_n | p \rangle \langle p' | E_{n'} \rangle \underline{F_{pp'}}$$

(b) $F_{pp'} \to F_{nn'}$

$$\underline{F_{pp'}} = \langle p | \hat{F} | p' \rangle = \sum_n \sum_{n'} \langle p | E_n \rangle \langle E_n | \hat{F} | E_{n'} \rangle \langle E_{n'} | p' \rangle$$

$$= \sum_n \sum_{n'} \langle p | E_n \rangle \langle E_{n'} | p' \rangle \underline{F_{nn'}}$$

§1.4 一维线性谐振子的代数解法

- $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$ ，（哈密顿算符）

- $\hat{H}|E\rangle = E|E\rangle$ ，（本征值方程）

- 引入算符 $\hat{a}$ 及其厄米共轭算符 $\hat{a}^\dagger$

（线性算符）

$$\begin{cases} \hat{a} \equiv \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \\ \hat{a}^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right). \end{cases}$$

* $\hat{a}$ 和 $\hat{a}^\dagger$ 都不是厄米算符.

$$\hat{a}^\dagger \neq \hat{a} .$$

- $\hat{a}$ 与 $\hat{a}^+$ 的对易关系

$$[\hat{a}, \hat{a}^+] = \left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\right],$$

利用对易关系：$[\hat{x}, \hat{p}] = i\hbar$，$[\hat{x}, \hat{x}] = [\hat{p}, \hat{p}] = 0$，

可得：$$\boxed{[\hat{a}, \hat{a}^+] = 1}$$

- 逆变换

$$\begin{cases} \hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^+ + \hat{a}\right) \\ \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a}^+ - \hat{a}\right) \end{cases} \xrightarrow{\text{代入 } \hat{H} \text{ 中}}$$

$$\boxed{\hat{H} = \left(\hat{a}^+\hat{a} + \frac{1}{2}\right)\hbar\omega}$$

* $\hat{a}^+\hat{a}$ 是厄米算符.

定义：$\hat{a}^{+}\hat{a} \equiv \hat{N}$ （粒子数算符 / 占据算符）

$\Rightarrow$ $\boxed{\hat{H} = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega}$

- $[\hat{N}, \hat{H}] = 0$.

共同本征矢：$|n\rangle$

$$\begin{cases} \hat{N}|n\rangle = n|n\rangle \\ \hat{H}|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle \end{cases}$$

本征值方程：$\hat{N}|n\rangle = n|n\rangle$ （$|n\rangle \neq$ 零矢，$\langle n|n\rangle = 1$）

同时求 n 及 $|n\rangle$.

※ $\hat{N}$ 的本征谱的特性.

(i) n 是实数 （$\hat{N}$ 为厄密算符）

· 用 $\langle n|$ 左乘本征值方程两边，

$$\Rightarrow \quad \langle n|\hat{N}|n\rangle = n\langle n|n\rangle = n$$

$$\Rightarrow \quad \langle n|\hat{a}^{\dagger}\hat{a}|n\rangle = n$$

回顾：$\langle \psi|\psi\rangle \geq 0$

· 仅当 $|\psi\rangle = |$零$\rangle$ 时等式成立！

定义，$\hat{a}|n\rangle \equiv |\psi_1\rangle$ ，则 $\langle n|\hat{a}^\dagger = \langle\psi_1|$ ，

$$\langle n|\underbrace{\hat{a}^\dagger\hat{a}}_{\hat{N}}|n\rangle = n \quad \Rightarrow \quad \langle\psi_1|\psi_1\rangle = n \ ,$$

由 $\langle\psi_1|\psi_1\rangle \geq 0 \quad \Rightarrow \quad \underline{n \geqslant 0}$

(ii) n必须是包括零的正实数

(a) 当n取最小值($n=0$)时，必有 ($\because \langle\psi_1|\psi_1\rangle = 0$)

$$|\psi_1\rangle = |\text{零}\rangle \ , \ \langle\psi_1| = \langle\text{零}|$$

$$\Rightarrow \quad \underline{\hat{a}|0\rangle = |\text{零}\rangle} \ , \ \underline{\langle 0|\hat{a}^\dagger = \langle\text{零}|} .$$

(b) $\langle n|(\hat{a}^\dagger)^m(\hat{a})^m|n\rangle = ?$

厄密算符

(m为大于零的正整数)

- $m=2$ 情况

$$(\hat{a}^\dagger)^2(\hat{a})^2 = \hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}$$

$$= \hat{a}^\dagger(\hat{a}\hat{a}^\dagger - 1)\hat{a}$$

$$= \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a}$$

$$= \hat{N}^2 - \hat{N} = \hat{N}(\hat{N}-1).$$

$$\Rightarrow \quad \langle n|(\hat{a}^\dagger)^2(\hat{a})^2|n\rangle$$

$$= \langle n|\hat{N}(\hat{N}-1)|n\rangle = n(n-1).$$

定义 $(\hat{a}^2)|n\rangle \equiv |\psi_2\rangle$, $\langle n|(\hat{a}^\dagger)^2 \equiv \langle\psi_2|$

$$\Rightarrow \quad \langle\psi_2|\psi_2\rangle = n(n-1) \geq 0$$

(iii) n 必须是 0, 1 或大于 1 的正实数

禁区： $0 < n < 1$ $\left(\hat{a}^2|1\rangle = |零\rangle\right)$

- $m = 3$ 情況

- $(\hat{a}^+)^3(\hat{a})^3 = \hat{N}(\hat{N}-1)(\hat{N}-2)$

$$\Rightarrow \quad \langle n|(\hat{a}^+)^3(\hat{a})^3|n\rangle = \langle n|\hat{N}(\hat{N}-1)(\hat{N}-2)|n\rangle$$

$$= n(n-1)(n-2)$$

定义 $(\hat{a})^3|n\rangle \equiv |\psi_3\rangle$ 则

$$\langle\psi_3|\psi_3\rangle = n(n-1)(n-2) \geq 0$$

(iv) $n = 0, 1, 2,$ 或 $n > 2$ 的正实数

禁区: $0 < n < 1$, $1 < n < 2$

($\hat{a}^3|2\rangle = |零\rangle$)

$$\vdots$$

$$(\hat{a}^+)^m (\hat{a})^m =$$

$$\hat{N}(\hat{N}-1)(\hat{N}-2)\cdots(\hat{N}-(m-1))$$

$$\Rightarrow \quad \langle n|(\hat{a}^+)^m(\hat{a})^m|n\rangle$$

$$= n(n-1)(n-2)\cdots(n-(m-1)) \geq 0,$$

讨论 $m = 1, 2, 3, \cdots\cdots \infty$ 情况 $\Rightarrow$

(V) $\hat{N}$ 的本征值 n 所可取值为

$$n = 0, 1, 2, \cdots \infty$$

包括零的正整数

$$\hat{N}|n\rangle = n|n\rangle$$

$$n = 0, 1, 2, \cdots \infty$$

$$\left(\hat{a}^{n+1}|n\rangle = |零\rangle\right)$$

- 本征矢 $|n\rangle$ 的表达式

$$[\hat{N}, \hat{a}^+] = [\hat{a}^+\hat{a}, \hat{a}^+]$$

$$= \hat{a}^+[\hat{a}, \hat{a}^+] + [\hat{a}^+, \hat{a}^+]\hat{a}$$

$$= \hat{a}^+[\hat{a}, \hat{a}^+] = \hat{a}^+ ,$$

$$\Rightarrow \quad \hat{N}\hat{a}^+ = \hat{a}^+(\hat{N}+1)$$

$$\Rightarrow \quad \hat{N}(\hat{a}^+)^m = (\hat{a}^+)^m(\hat{N}+m) \qquad (m\text{为正整数})$$

作用于 $\hat{N}$ 的最小本征值 $n=0$ 对应的态 $|0\rangle$ 上,

$$\Rightarrow \quad \hat{N}(\hat{a}^+)^m|0\rangle = (\hat{a}^+)^m(\hat{N}+m)|0\rangle$$

$$= (\hat{a}^+)^m(0+m)|0\rangle$$

$$= m\,(\hat{a}^+)^m|0\rangle$$

$\Rightarrow$ $(\hat{a}^+)^m|0\rangle$ 是 $\hat{N}$ 的本征值为 m 的本征态!

$$|n\rangle \propto (\hat{a}^+)^n|0\rangle$$

$$|n\rangle = A(\hat{a}^+)^n|0\rangle$$

A 由 $\langle n|n\rangle = 1$ 确定!

· $\langle n| = A^* \langle 0|(\hat{a})^n$

$\Rightarrow \quad \langle n|n\rangle = |A|^2 \langle 0|(\hat{a})^n(\hat{a}^+)^n|0\rangle$

· 利用关系式：

$$[\hat{a}, \hat{f}(\hat{a}^+, \hat{a})] = \frac{\partial}{\partial \hat{a}^+}\hat{f}(\hat{a}^+, \hat{a})$$

(其中 $\hat{f}$ 为 $\hat{a}^+, \hat{a}$ 的零级数)

易得：$(\hat{a})^n(\hat{a}^+)^n|0\rangle = n!|0\rangle$

推导过程中利用了关系式

$$\hat{a}|0\rangle = |零\rangle$$

$$\Rightarrow \quad \langle n|n\rangle = |A|^2 n!$$

$$\Rightarrow \quad |A|^2 n! = 1 \qquad (取A为实数)$$

$$\Rightarrow \quad A = \frac{1}{\sqrt{n!}}$$

$$\Rightarrow \quad \boxed{|n\rangle = \frac{(\hat{a}^+)^n}{\sqrt{n!}}|0\rangle}$$

$$\hat{a}^+|n\rangle = \frac{\hat{a}^+(\hat{a}^+)^n}{\sqrt{n!}}|0\rangle$$

$$= \frac{\sqrt{n+1}(\hat{a}^+)^{n+1}}{\sqrt{(n+1)!}}|0\rangle$$

$$= \sqrt{n+1}\,|n+1\rangle \qquad (上升算符)$$

$$\cdot\ \hat{a}|n\rangle = \hat{a}\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$$

$$\because\quad \hat{a}(\hat{a}^\dagger)^n = (\hat{a}^\dagger)^n\hat{a} + n(\hat{a}^\dagger)^{n-1} \qquad (\hat{f} = (\hat{a}^\dagger)^n)$$

$$\cdot\ \hat{a}|n\rangle = \frac{\hat{a}(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle = \frac{(\hat{a}^\dagger)^n\hat{a}|0\rangle}{\sqrt{n!}} + \frac{n(\hat{a}^\dagger)^{n-1}|0\rangle}{\sqrt{n!}}$$

$(n>0)$ $\qquad\qquad$ $=0$

$$= \frac{n}{\sqrt{n}}\frac{(\hat{a}^\dagger)^{n-1}}{\sqrt{(n-1)!}}|0\rangle$$

$$= \sqrt{n}\,|n-1\rangle \qquad (\text{下降算符})$$

$$\cdot\ \hat{N}|n\rangle = \hat{a}^\dagger\hat{a}\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle = \sqrt{n}\,\hat{a}^\dagger|n-1\rangle$$

$$= \sqrt{n}\sqrt{n}\,|n\rangle$$

$$= n|n\rangle$$

(OK!)

· $(\hat{a})^{n+1}|n\rangle \neq |零\rangle$.

$$(\hat{a})^{n+1}|n\rangle = \hat{a}\ \hat{a}^n|n\rangle$$

$$\propto \hat{a}\,|n-n\rangle$$

$$= \hat{a}|0\rangle = |零\rangle$$